



Earth Magnetic Field Navigation Algorithms for Aerospace Vehicles

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2^{ème} année

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Thèse financée par	ONERA	

Mots clés : Navigation, Filtrage particulaire, Krigeage, SLAM, Magnétomètres

Innovative navigation algorithms utilizing Earth's magnetic field are being developed to support aerospace applications without relying on GNSS. While inertial navigation systems are reliable for short missions, their accuracy diminishes over time due to biases and noise. To overcome this, external recalibration methods, typically dependent on GNSS, can be used. However, magnetic field data is more reliable as it is available regardless of location or time.

Magnetic field-based navigation offers significant advantages thanks to contributions in magnetometer technology and calibration techniques. Statistical methods like kriging enable real-time reconstruction of magnetic field maps, even in high-altitude areas with weak magnetic signals. Particle filtering techniques handle non-linearities and ambiguities in data, enhancing the robustness of the navigation system, and can be combined with kriging using a SLAM (Simultaneous Localisation and Mapping) based framework as shown by figure 1.

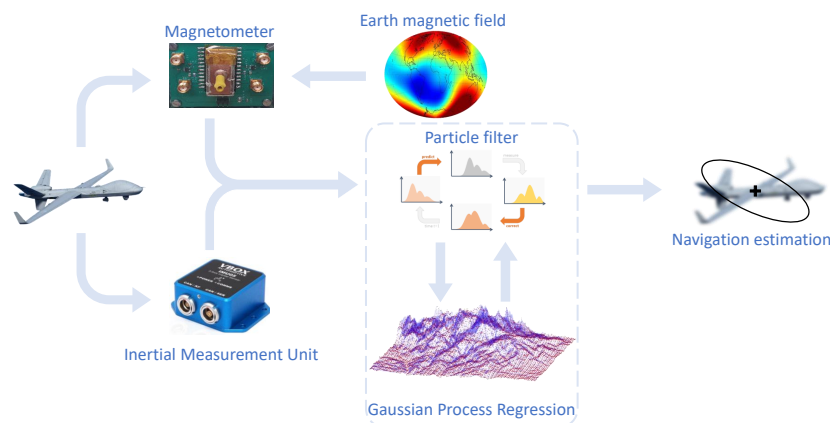


Figure 1: Structure of a magnetic field aided SLAM for autonomous navigation

A new method, the Adaptive Kriging Particle Filter, has been introduced to optimise computational efficiency, showing superior performance in accuracy and convergence compared to traditional approaches. Future work focuses on topological data selection, improving error estimation, and integrating SLAM frameworks to create adaptable and reliable navigation solutions for complex aerospace missions, as well as expanding this method to other physical fields such as gravitational and altimetric navigation.